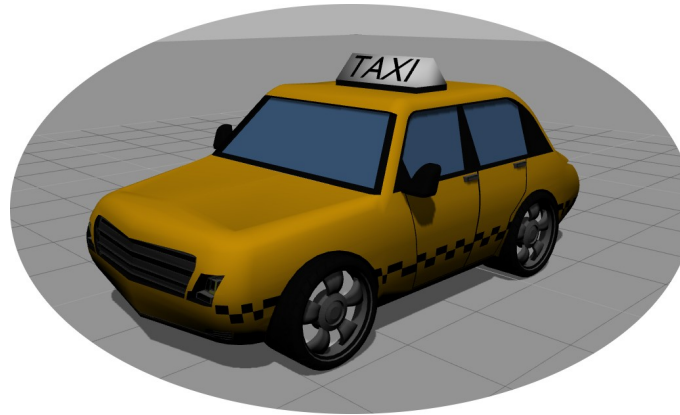




***Robots
sociales***

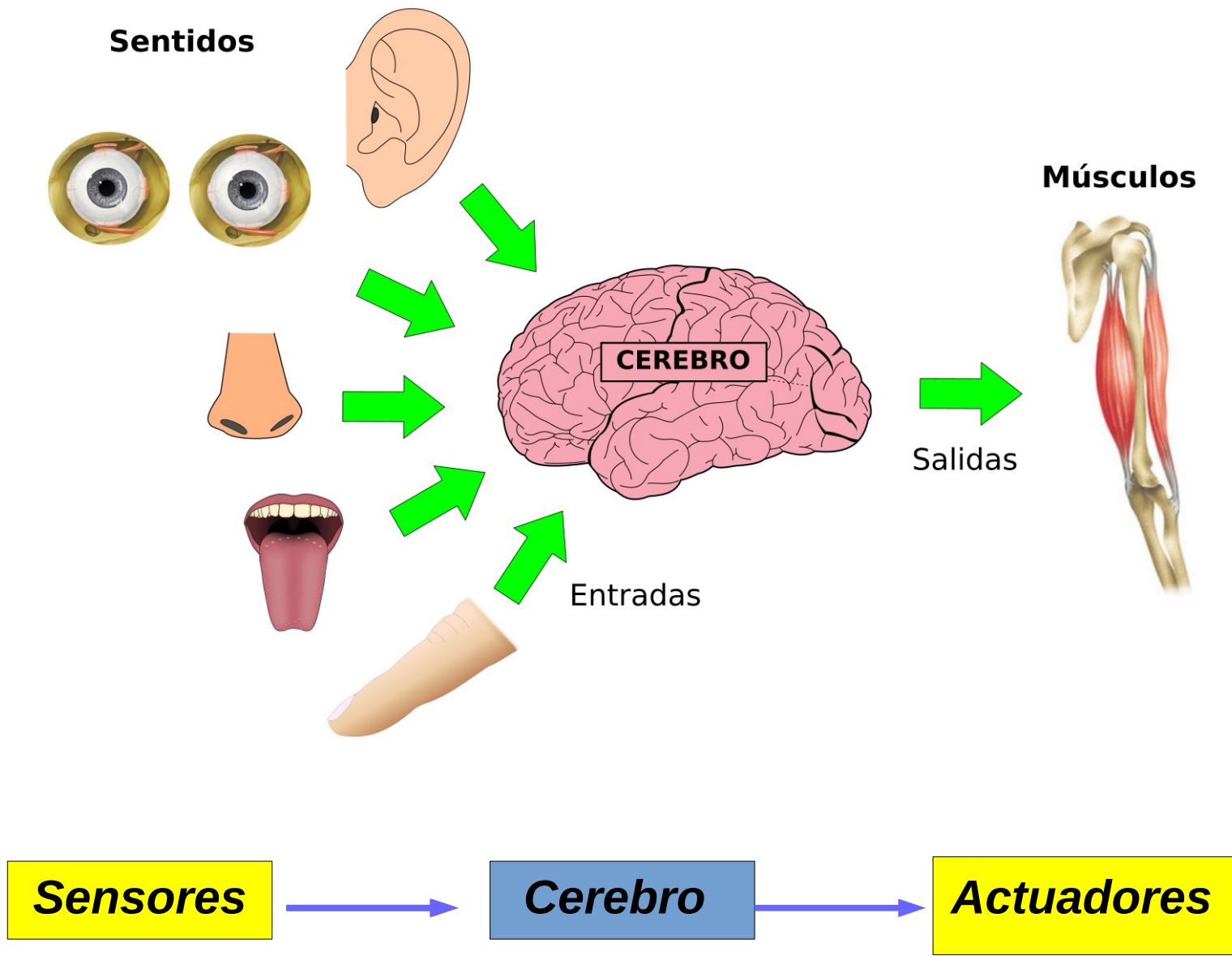


***Vehículos
autónomos***



Drones

Componentes de un robot



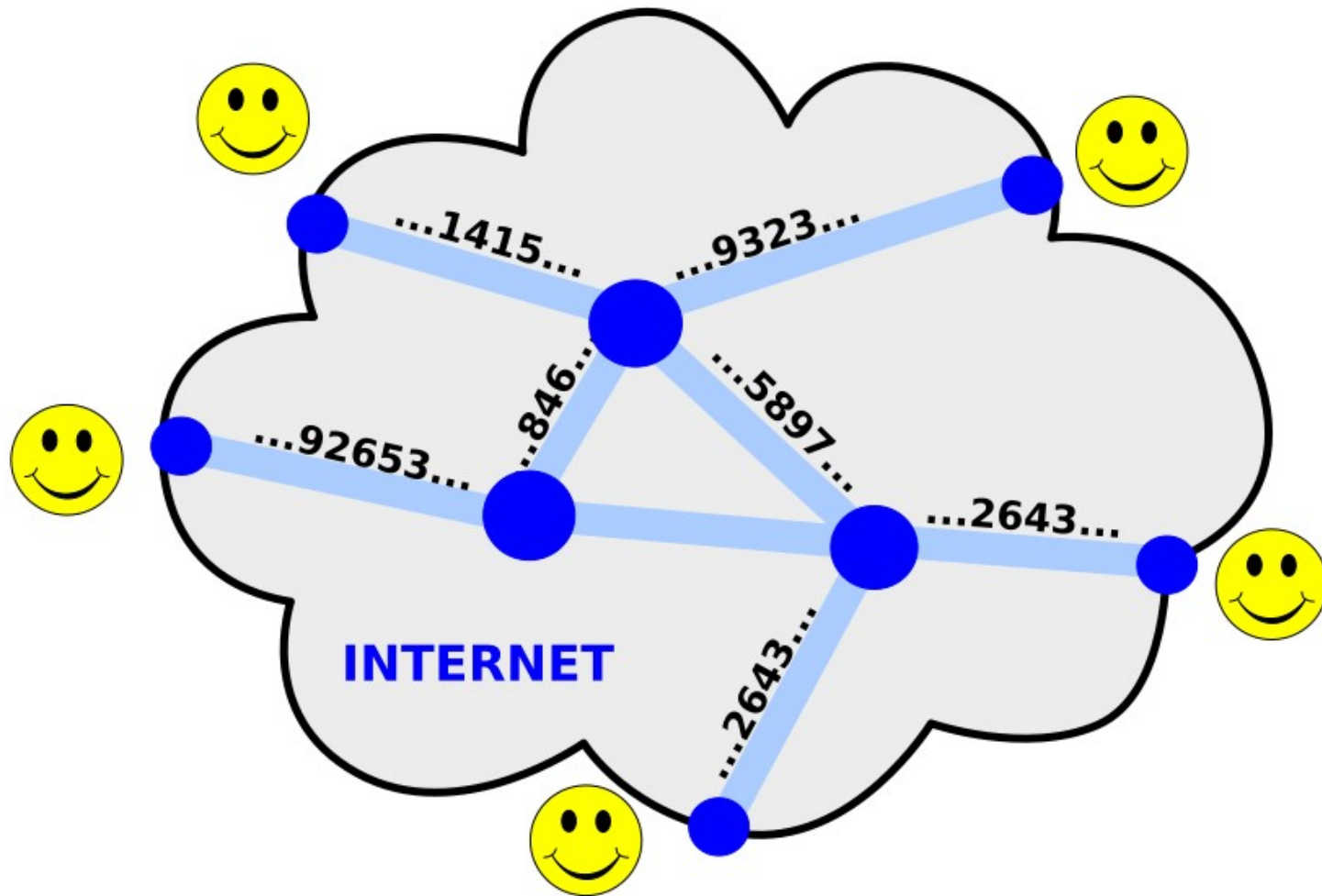
Mundo digital



1415926535 8979323846 2643383279
5028841971 6939937510 5820974944
5923078164 0628620899 8628034825
3421170679 8214808651 3282306647
7245870066 0631558817 4881520920
9628292540 9171536436 7892590360
0113305305 4882046652 1384146951
9415116094 3305727036 5759591953
0921861173 8193261179 3105118548
0744623799 6274956735 1885752724
8912279381 8301194912 9833673362
4406566430 8602139494 6395224737
1907021798 6094370277 0539217176
2931767523 8467481846 7669405132

TODO SON NÚMEROS

Internet

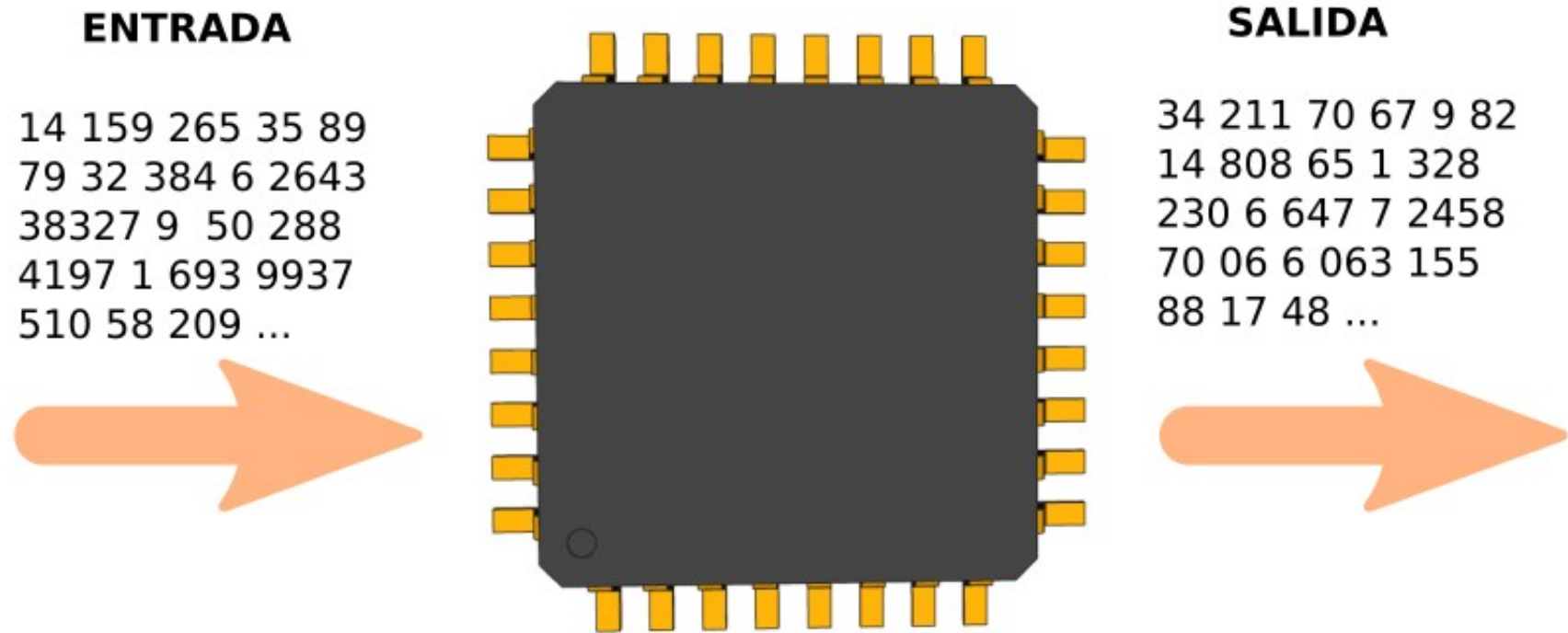


*Transporte, almacenamiento y manipulación
de NÚMEROS*

Los pitagóricos estarían orgullosos



Electrónica digital (I)



*Circuitos electrónicos que manipulan,
transportan y almacenan NÚMEROS*

Bits

14159265358979323846264
33832795028841971693993
75105820974944592307816

Números en decimal



0001 0100 0001 0101 1001 0010 0110 0101 0011
0101 1000 1001 0111 1001 0101 0000 0010 1000
1000 0100 0001 0001 1001 0111 0001 0110 0111

Digitos Binarios

TODO se reduce a 1s y 0s

0 1

Ser o no ser

Par o impar

Verdadero o falso

Todo o nada

Yin y Yang

Culo o codo...

Experimentando: Examinemos una imagen



1200px-Bronnikov_gimnpifagoreizev.jpg - GHex

00000000	FF	D8	FF	E0	00	10	4A	46	49	46	00	01	01	01	00	48	00	48	00	00	FF
00000015	FE	00	54	46	69	6C	65	20	73	6F	75	72	63	65	3A	20	68	74	74	70	3A
0000002A	2F	2F	63	6F	6D	6D	6F	6E	73	2E	77	69	6B	69	6D	65	64	69	61	2E	6F
0000003F	72	67	2F	77	69	6B	69	2F	46	69	6C	65	3A	42	72	6F	6E	6E	69	6B	6F
00000054	76	5F	67	69	6D	6E	70	69	66	61	67	6F	72	65	69	7A	65	76	2E	6A	70
00000069	67	FF	E2	0C	58	49	43	43	5F	50	52	4F	46	49	4C	45	00	01	01	00	00
0000007E	0C	48	4C	69	6E	6F	02	10	00	00	6D	6E	74	72	52	47	42	20	58	59	5A
00000093	20	07	CE	00	02	00	09	00	06	00	31	00	00	61	63	73	70	4D	53	46	54
000000A8	00	00	00	00	49	45	43	20	73	52	47	42	00	00	00	00	00	00	00	00	00
000000BD	00	00	00	00	F6	D6	00	01	00	00	00	00	D3	2D	48	50	20	20	00	00	00
000000D2	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
000000E7	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
000000FC	00	00	00	11	63	70	72	74	00	00	01	50	00	00	00	33	64	65	73	63	00
00000111	00	01	84	00	00	00	6C	77	74	70	74	00	00	01	F0	00	00	00	14	62	6B
00000126	70	74	00	00	02	04	00	00	00	14	72	58	59	5A	00	00	02	18	00	00	00
0000013B	14	67	58	59	5A	00	00	02	2C	00	00	00	14	62	58	59	5A	00	00	02	40

.....JFIF.....H.H...
..TFile source: http:
//commons.wikimedia.o
rg/wiki/File:Bronniko
v_gimnpifagoreizev.jp
g...XICC_PROFILE.....
.HLino...mnrRGB XYZ
.....1..acspMSFT
....IEC sRGB.....
.....-HP ..
.....
.....
....cprt...P...3desc.
.....lwtpt.....bk
pt.....rXYZ.....
.gXYZ.....bXYZ...@

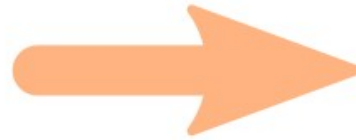
Signed 8 bit: -1 Signed 32 bit: -520103681 Hexadecimal: FF
Unsigned 8 bit: 255 Unsigned 32 bit: 3774863615 Octal: 377
Signed 16 bit: -9985 Signed 64 bit: 3774863615 Binary: 11111111
Unsigned 16 bit: 55551 Unsigned 64 bit: 3774863615 Stream Length: 8 - +
Float 32 bit: -1,474861e+20 Float 64 bit: 4,129770e+30
☒ Show little endian decoding ☐ Show unsigned and float as hexadecimal
Offset: 0x0

En el interior de los microchips

Productos electrónicos y circuitos

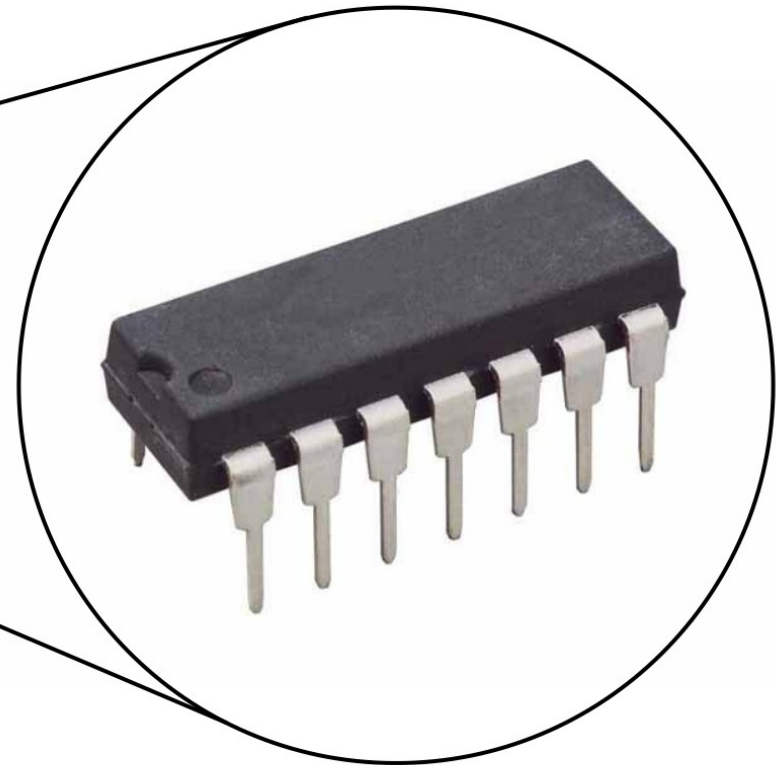
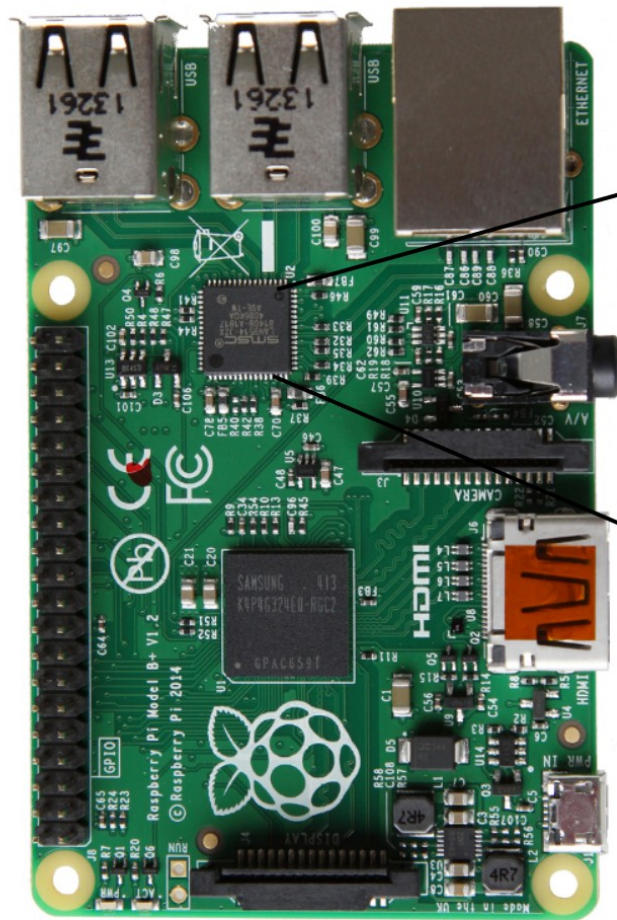


Producto Electrónico



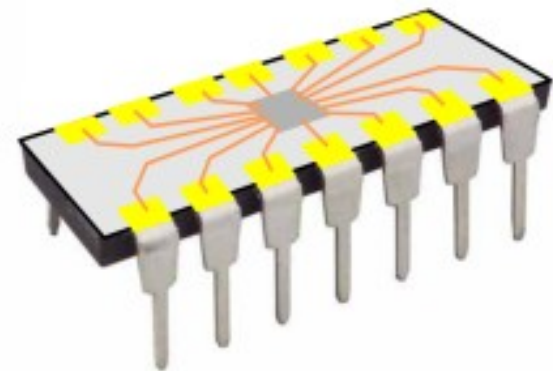
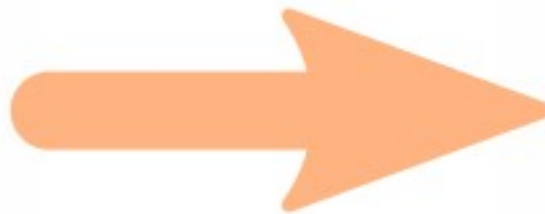
Circuito electrónico

PCBs y Circuitos integrados



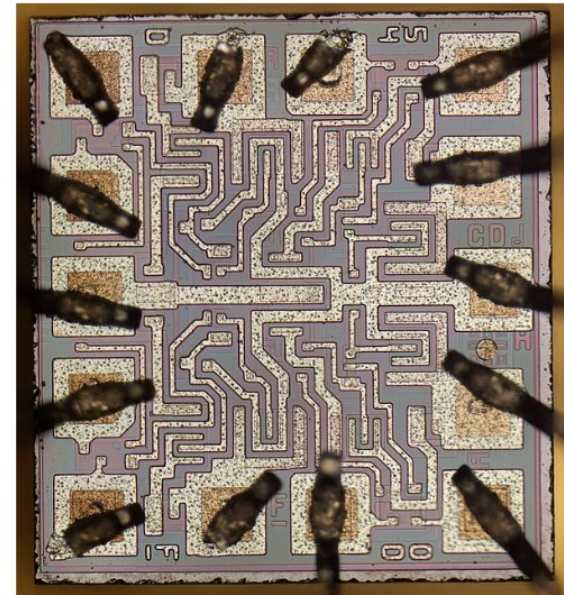
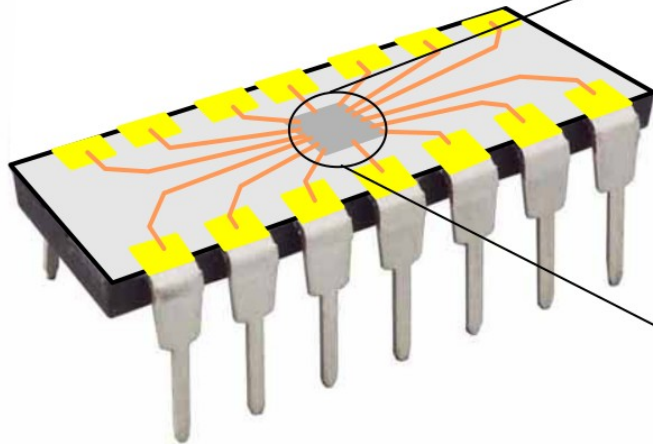
Encapsulado y dado

*Demo: Mostrar chips
Dip y eeprom*



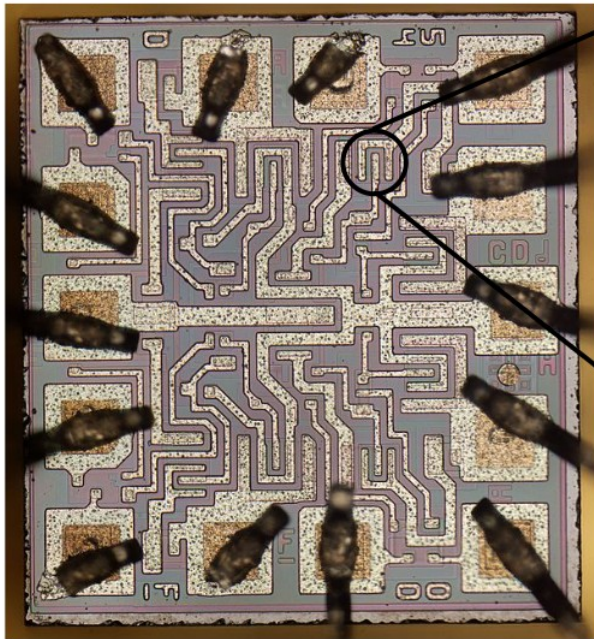
Dado de silicio

*Demo:
Proyecto 54/74*

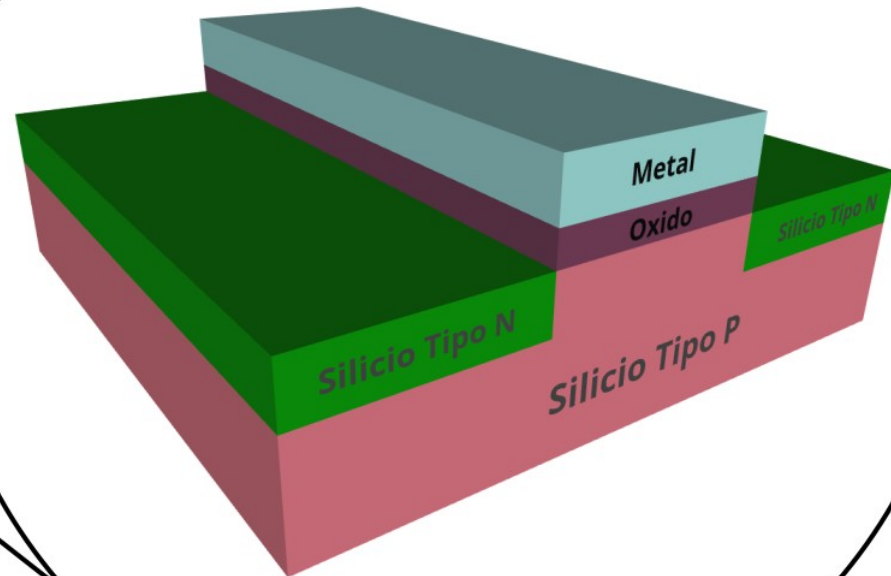


Semiconductores

Dado

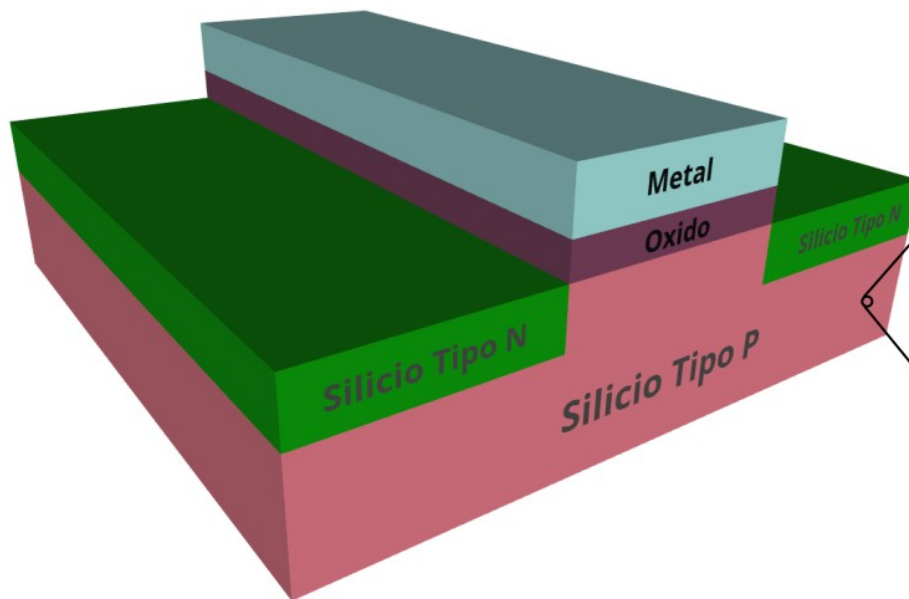


Transistor CMOS

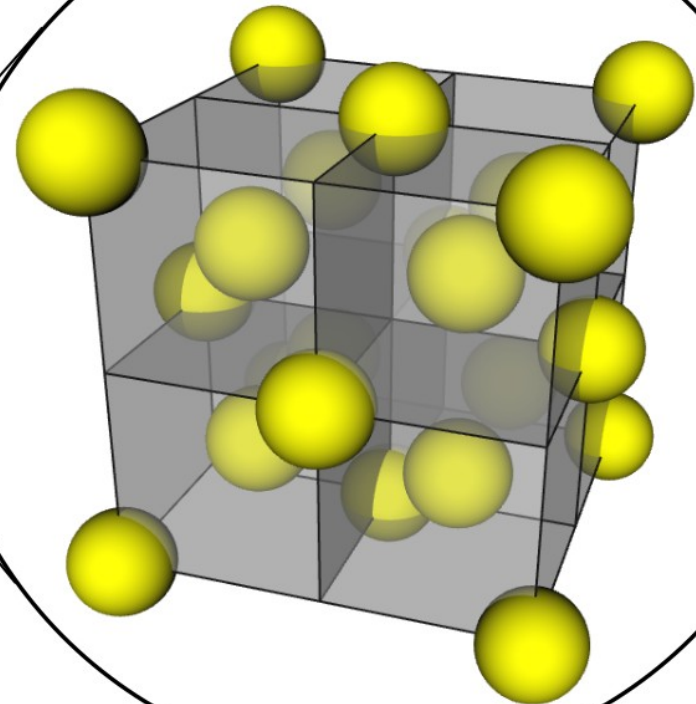


Cristal de silicio

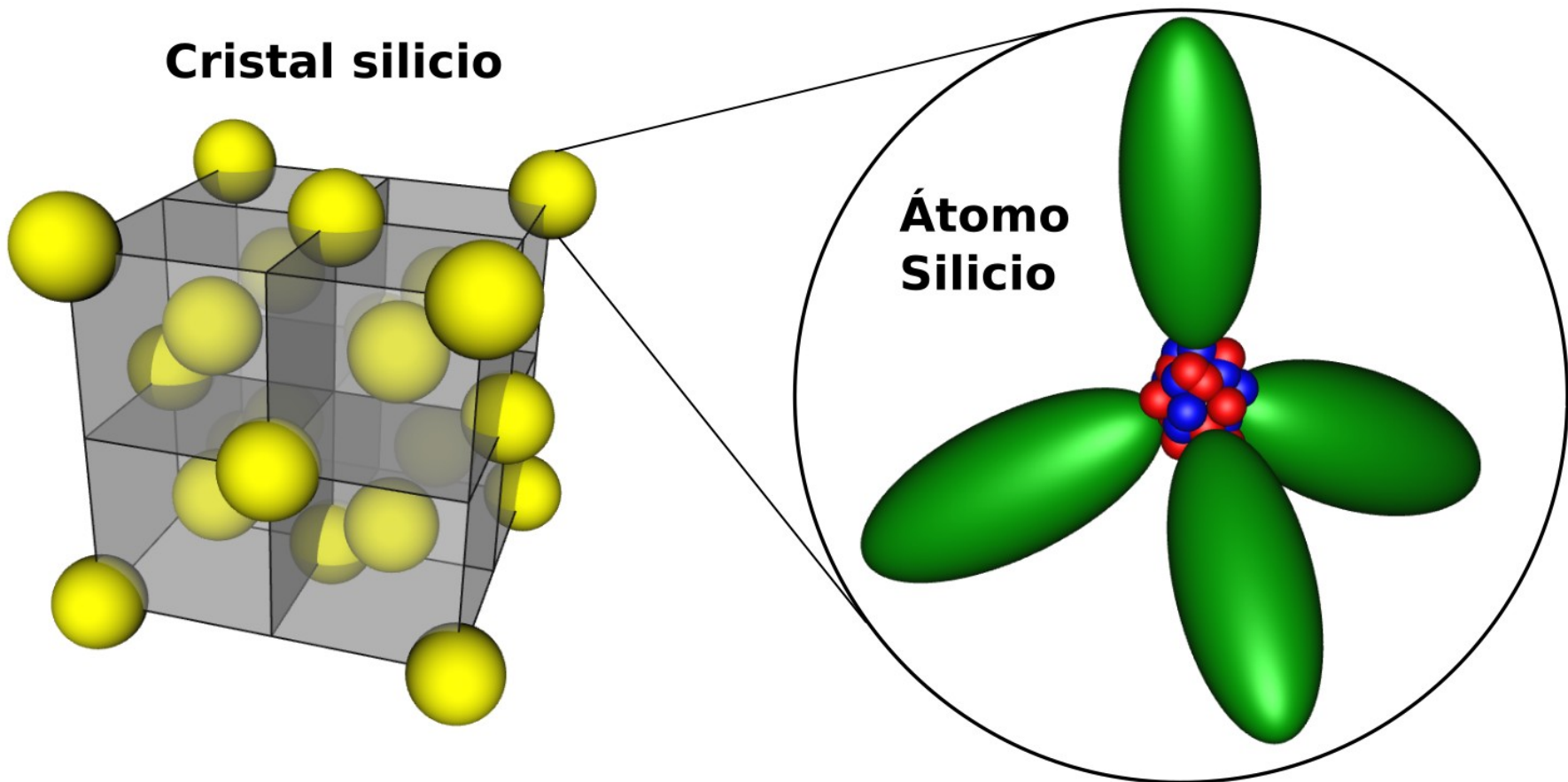
Transistor CMOS



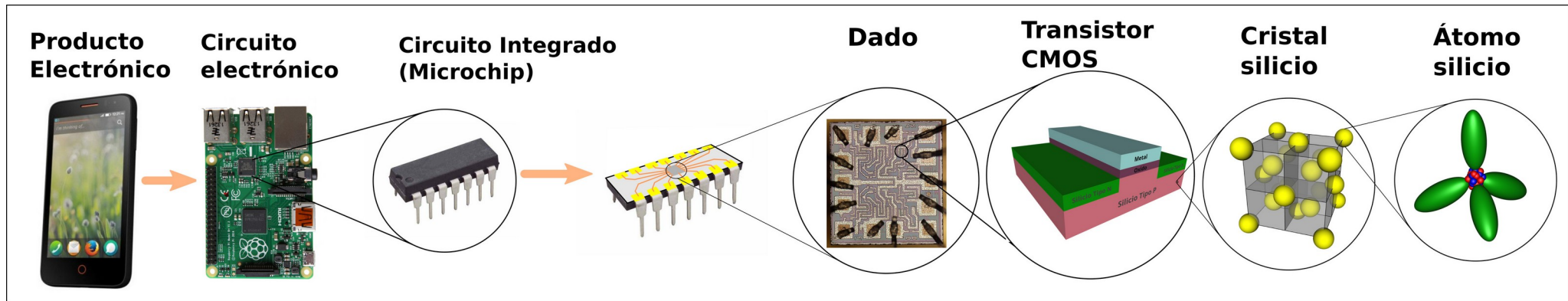
Cristal silicio



Átomos :-)

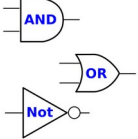
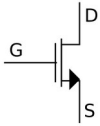
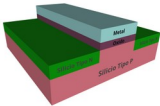
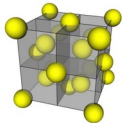
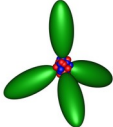


Del producto al átomo

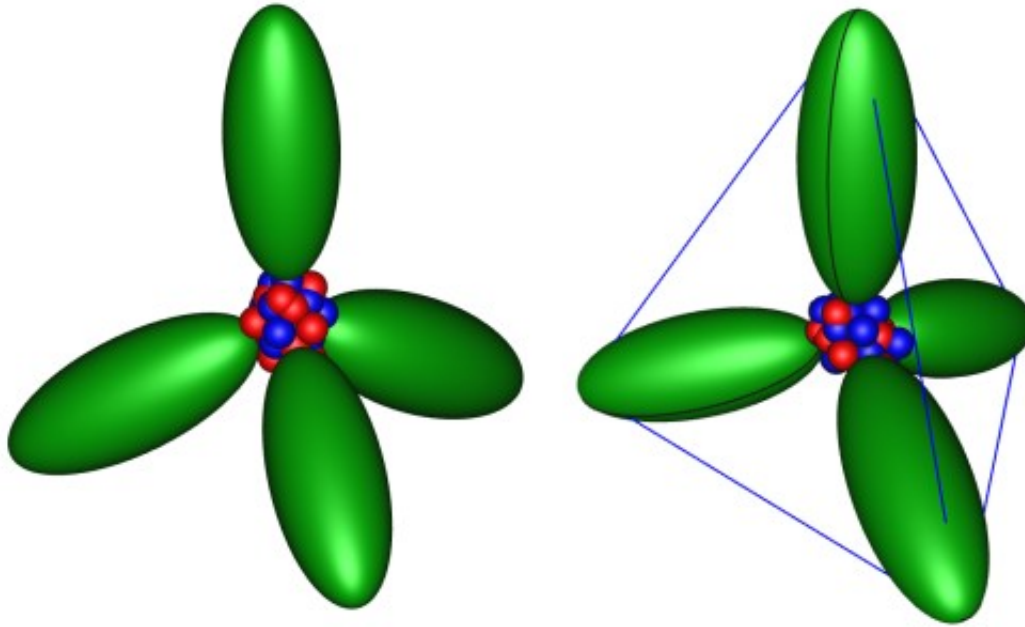


Niveles de descripción

Agrupación en niveles

	Nivel 7: Software
	Nivel 6: Microprocesador
	Nivel 5: Puertas lógicas
	Nivel 4: Transistor
	Nivel 3: Semiconductores
	Nivel 2: Materiales
	Nivel 1: Átomos

Nivel 1: Átomos

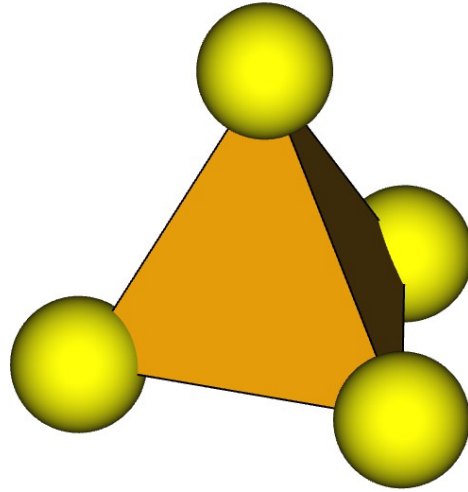
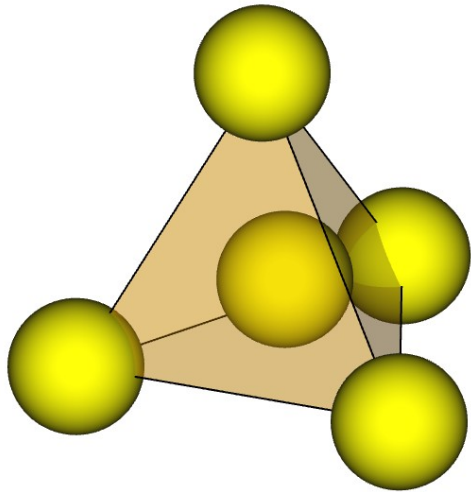


Átomo de silicio

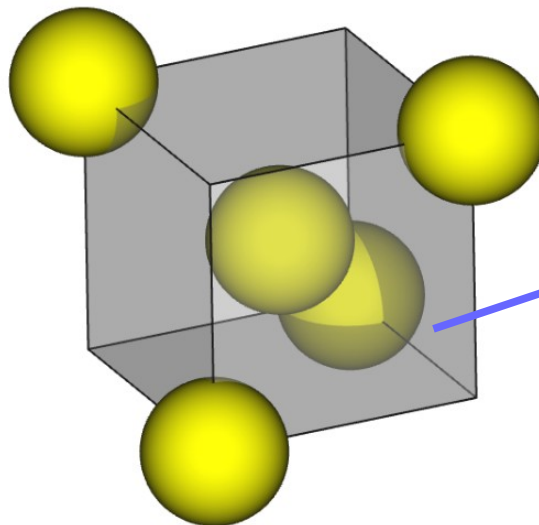
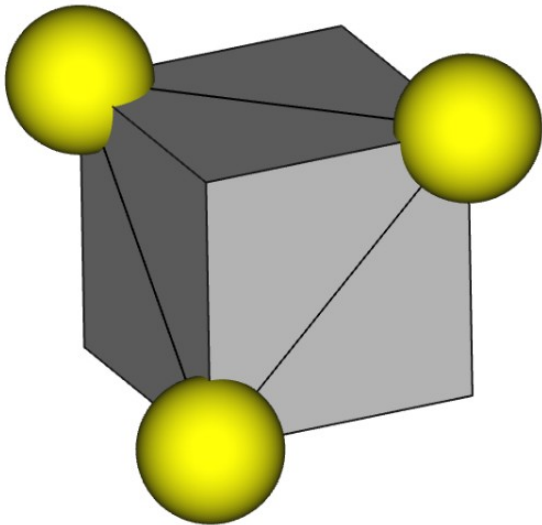
Física, Química

Experimentando: Abrir el átomo de Silicio en 3D con el programa FreeCAD

Nivel 2: Materiales



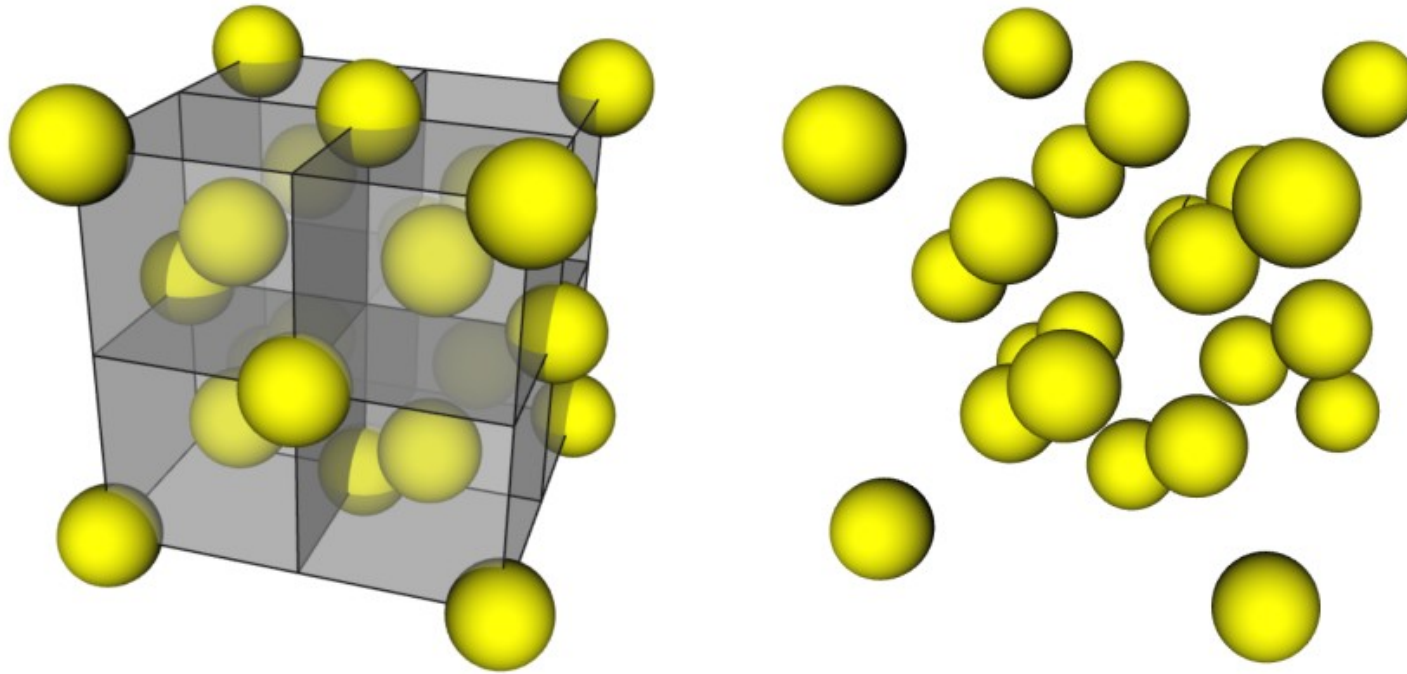
Cristal de silicio
Física del estado sólido



Celda de 5 átomos



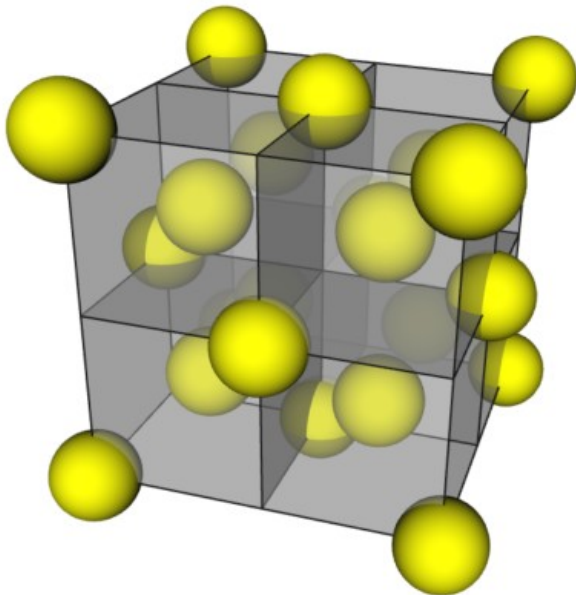
Nivel 2: Materiales



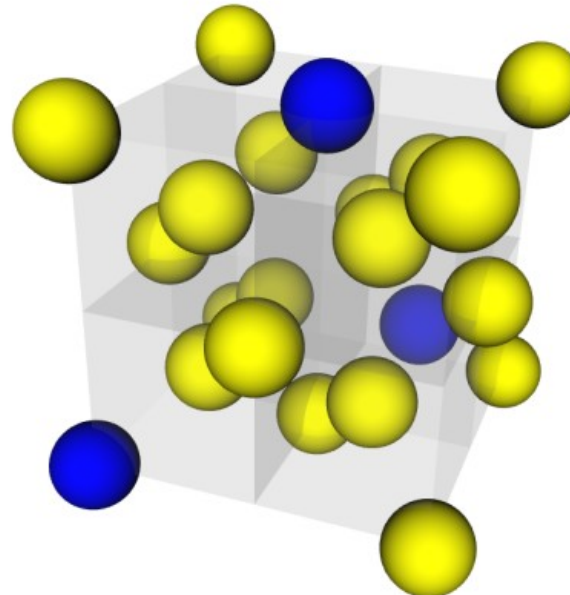
Experimentando: Abrir la celda de 5 átomos de Si y la estructura cristalina en FreeCAD

Nivel 3: Semiconductores

Cristal de Silicio puro



Cristal de Silicio dopado



Átomo de Silicio (Si)

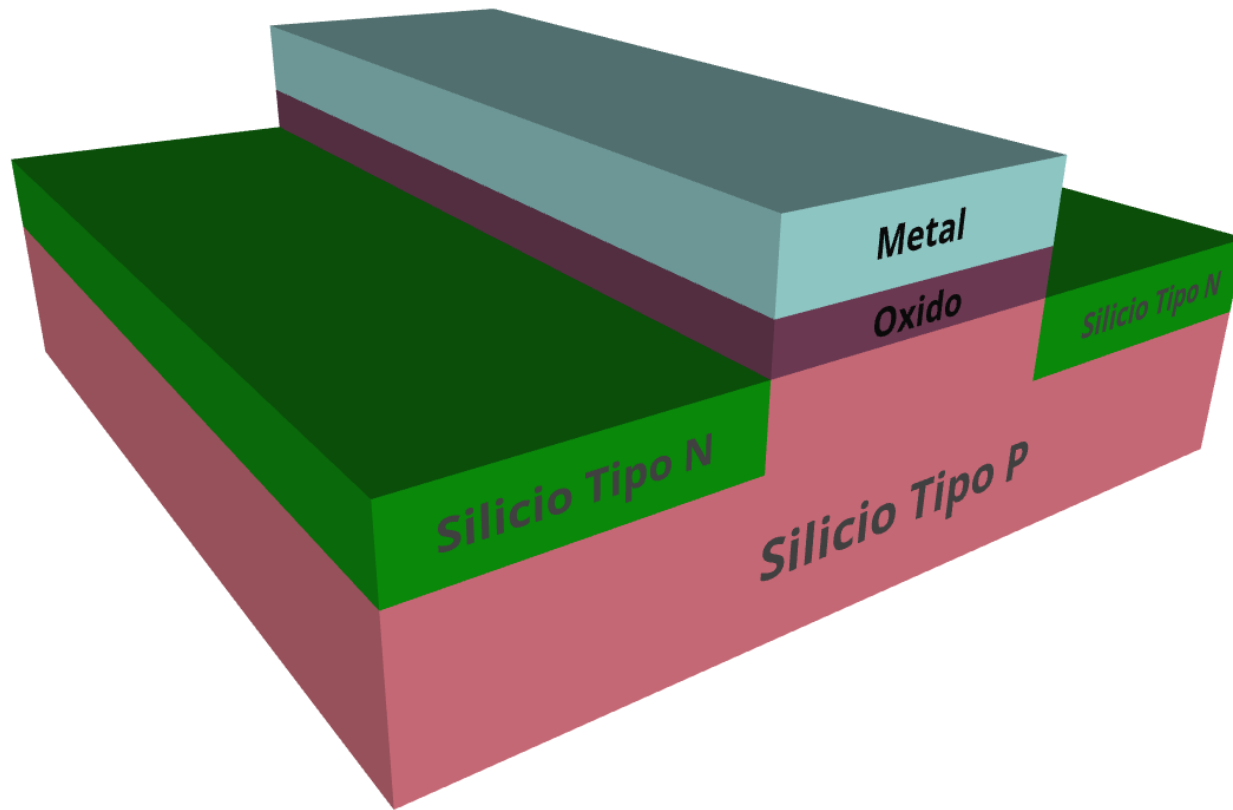


Átomo dopante (Ej. Fósforo, Boro)

Uniones PN

Electrónica de dispositivos

Nivel 3: Semiconductores



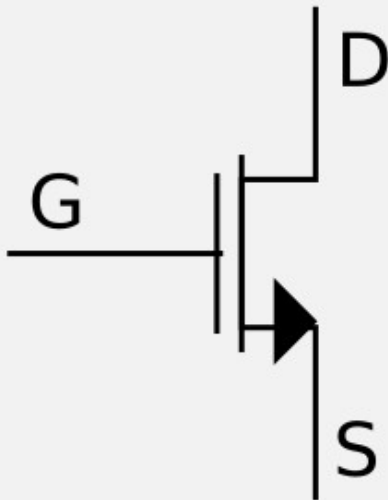
*Transistor
MOSFET*

Experimentando: Abrir el dibujo 3D de la estructura del MOSFET con FreeCAD

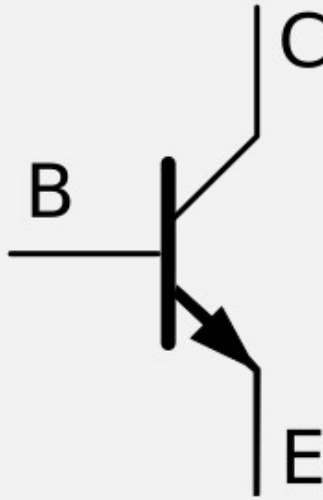
Nivel 4: Transistor

*Transistor
Microelectrónica*

**Transistor
MOSFET**

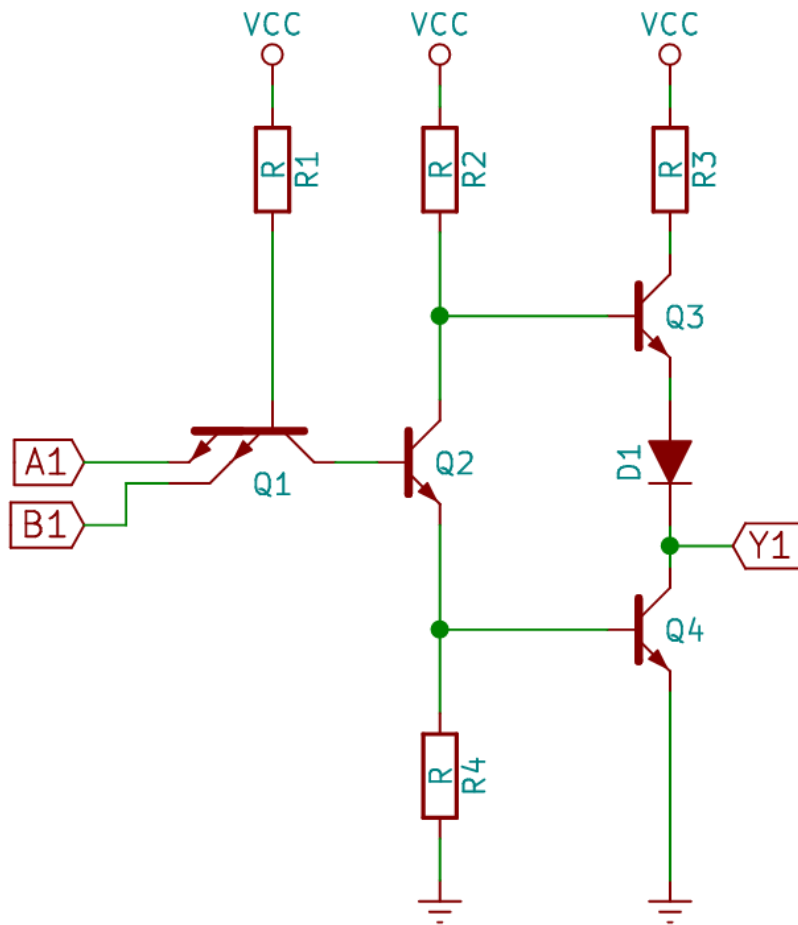


**Transistor
Bipolar (NPN)**



Nivel 4: Transistor

Implementación de una puerta NAND

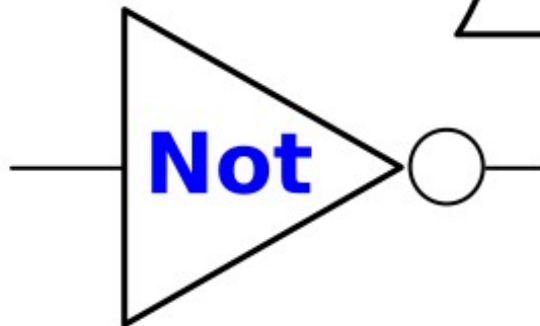
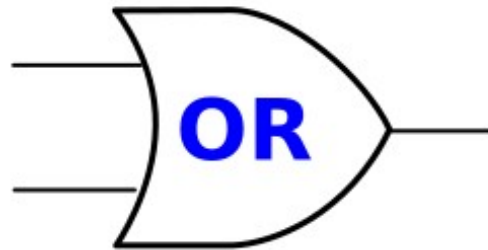
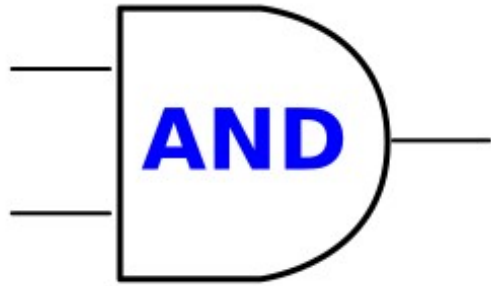


Experimentando: Abrir la puerta NAND del integrado 5400, con Kicad

Nivel 5: Puertas lógicas

Puertas Lógicas
Electrónica
Digital

¡Aparecen los bits!

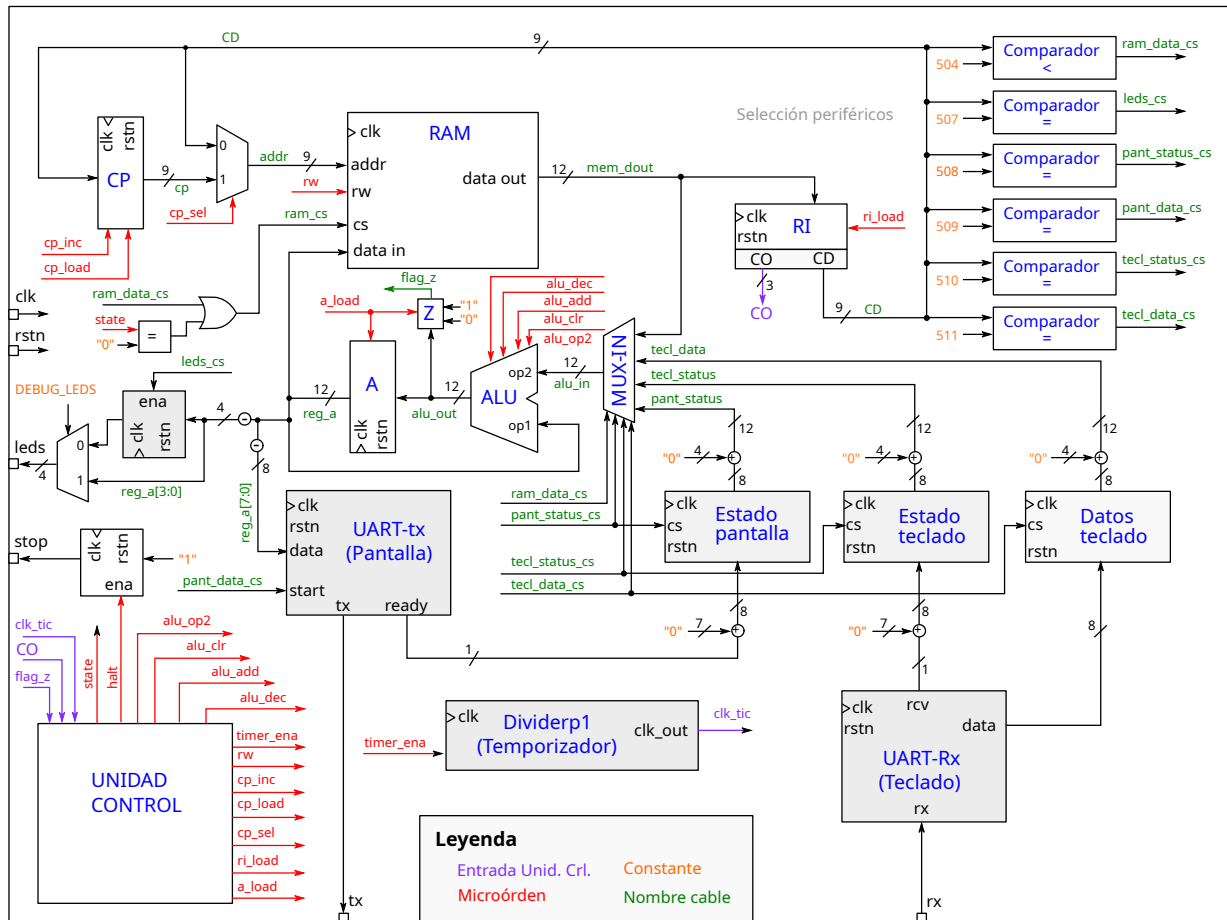


0 = 0 voltios

1 = 5 voltios

Nivel 6: Microprocesador

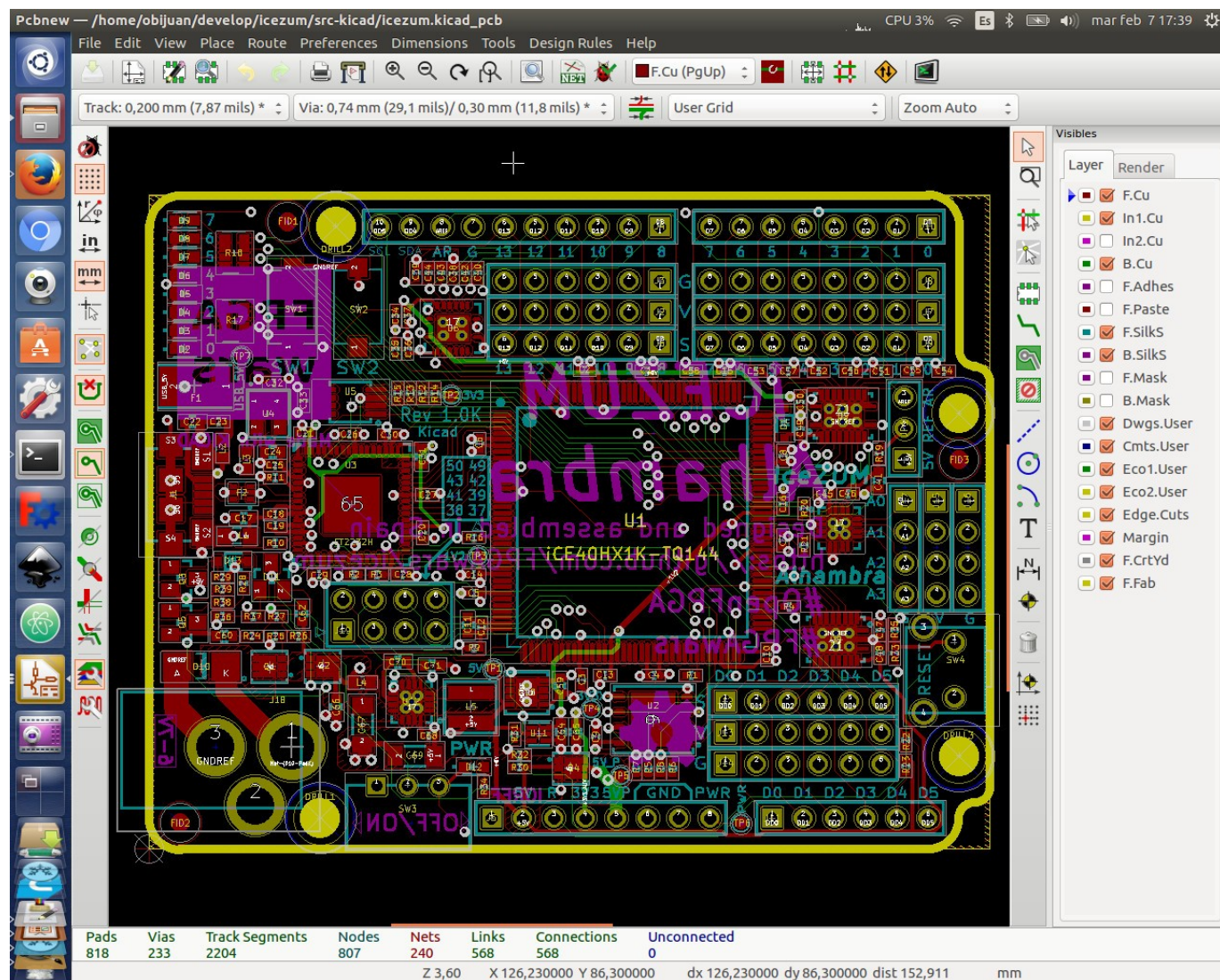
Microprocesador Arquitectura de Ordenadores



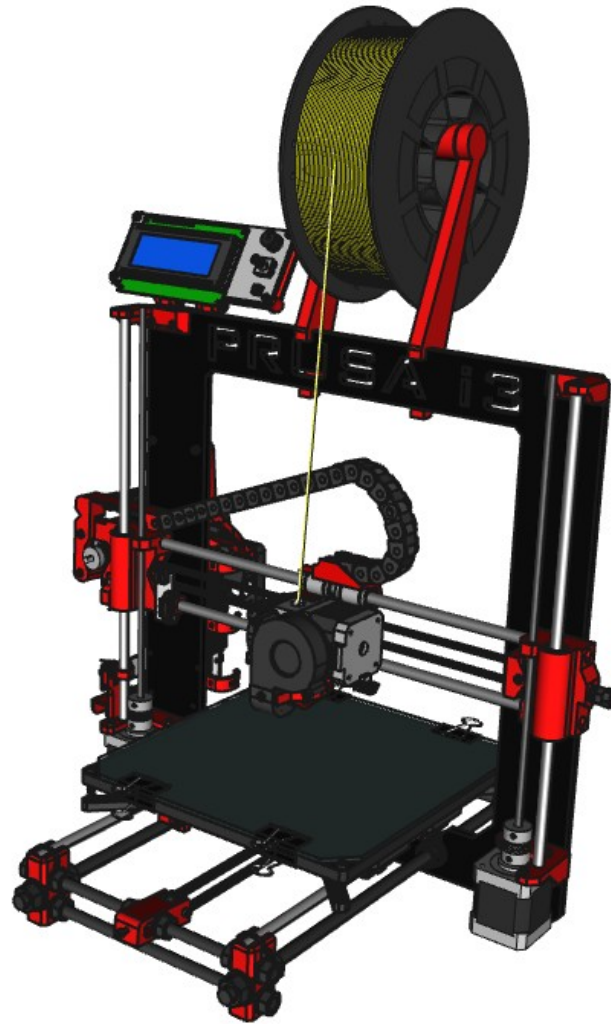
¡Máquina universal!

Máquina reprogramable

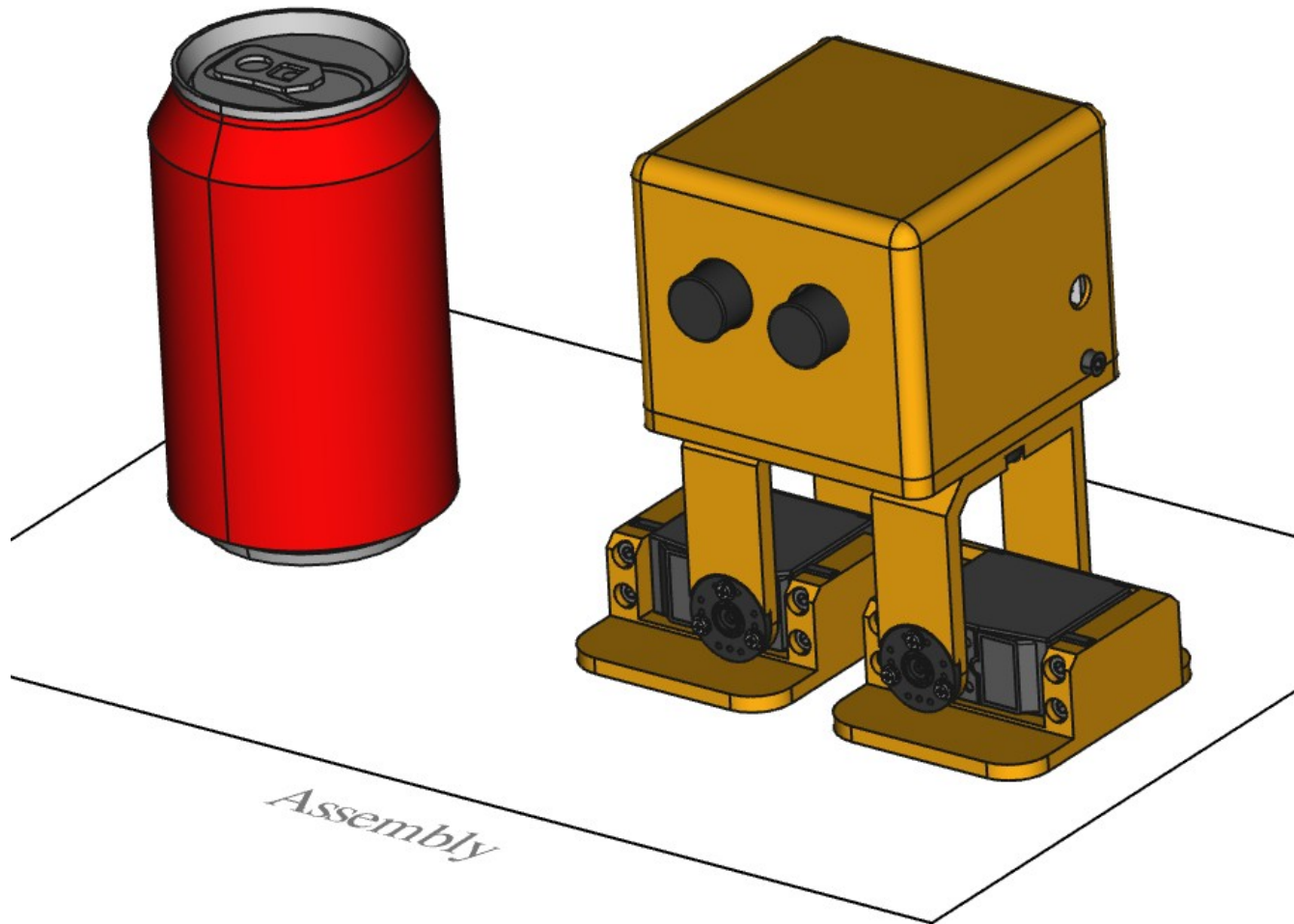
Placas



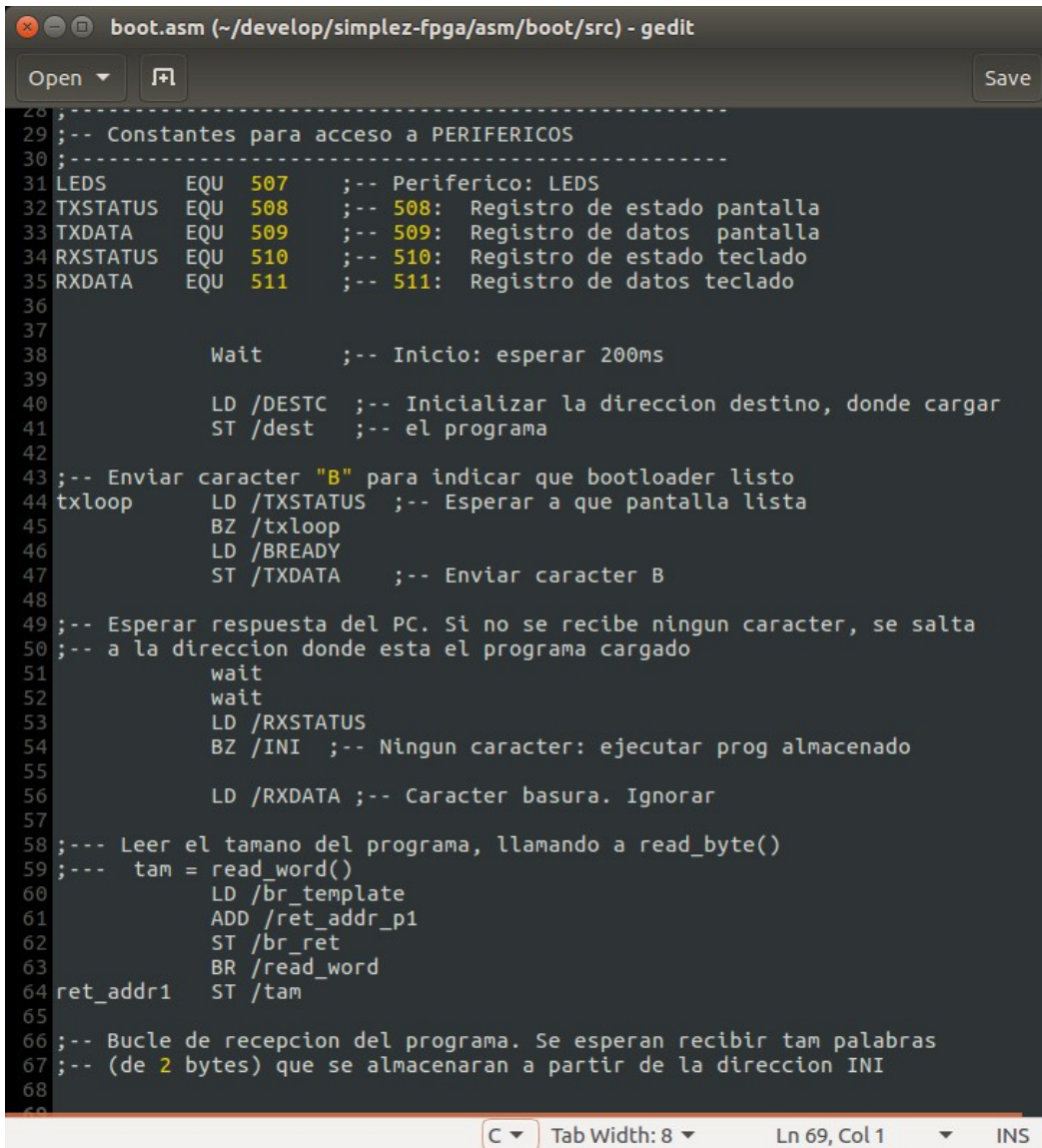
Estructura: Impresora 3D



Robots educativos



Nivel 7: Software



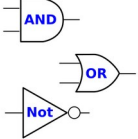
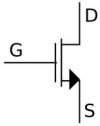
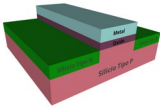
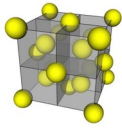
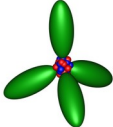
```
boot.asm (~/develop/simplez-fpga/asm/boot/src) - gedit
Open Save
28 ;-----
29 ;-- Constantes para acceso a PERIFERICOS
30 ;-----
31 LEDS      EQU 507    ;-- Periferico: LEDS
32 TXSTATUS  EQU 508    ;-- 508: Registro de estado pantalla
33 TXDATA    EQU 509    ;-- 509: Registro de datos pantalla
34 RXSTATUS  EQU 510    ;-- 510: Registro de estado teclado
35 RXDATA    EQU 511    ;-- 511: Registro de datos teclado
36
37
38      Wait      ;-- Inicio: esperar 200ms
39
40      LD /DESTC ;-- Inicializar la direccion destino, donde cargar
41      ST /dest  ;-- el programa
42
43 ;-- Enviar caracter "B" para indicar que bootloader listo
44 txloop  LD /TXSTATUS ;-- Esperar a que pantalla lista
45        BZ /txloop
46        LD /BREADY
47        ST /TXDATA ;-- Enviar caracter B
48
49 ;-- Esperar respuesta del PC. Si no se recibe ningun caracter, se salta
50 ;-- a la direccion donde esta el programa cargado
51      wait
52      wait
53      LD /RXSTATUS
54      BZ /INI ;-- Ningun caracter: ejecutar prog almacenado
55
56      LD /RXDATA ;-- Caracter basura. Ignorar
57
58 ;--- Leer el tamaño del programa, llamando a read_byte()
59 ;--- tam = read_word()
60      LD /br_template
61      ADD /ret_addr_p1
62      ST /br_ret
63      BR /read_word
64 ret_addr1 ST /tam
65
66 ;-- Bucle de recepcion del programa. Se esperan recibir tam palabras
67 ;-- (de 2 bytes) que se almacenaran a partir de la direccion INI
68
69
```

*Software
Programación*

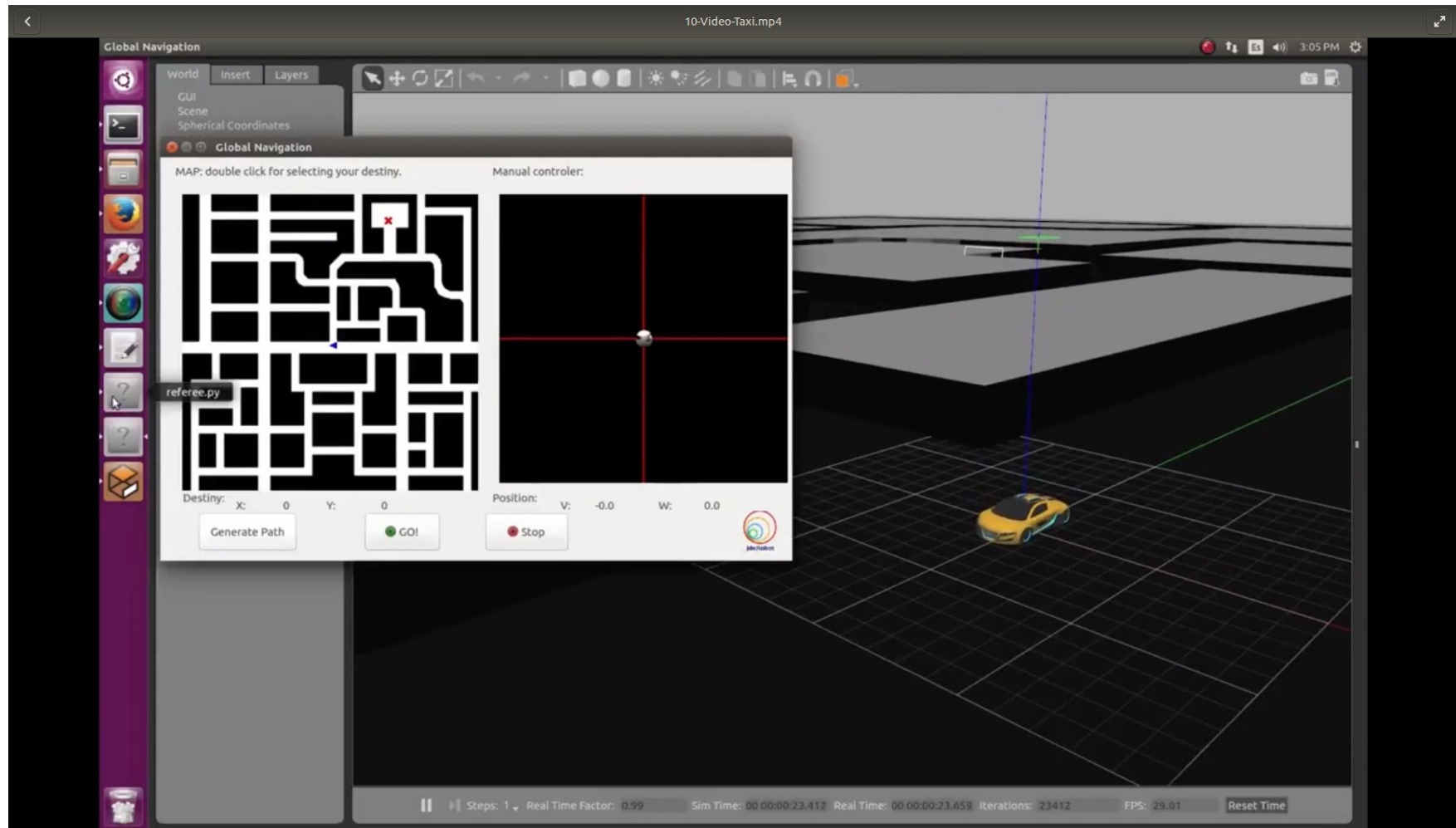
*Código
Algoritmos*

Muchos niveles de software

Agrupación en niveles

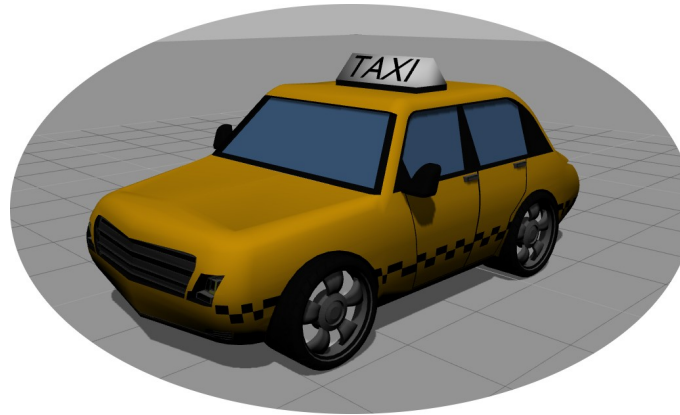
	Nivel 7: Software
	Nivel 6: Microprocesador
	Nivel 5: Puertas lógicas
	Nivel 4: Transistor
	Nivel 3: Semiconductores
	Nivel 2: Materiales
	Nivel 1: Átomos

Videos





***Robots
sociales***



***Vehículos
autónomos***



Drones